

A Holistic Approach to Assessing Quality across the Continuum of Healthcare by Aligning Provider-Centric Result Measures with Patient-Centric Value Measures

Facing the rigor of cost-cutting efforts that have driven efficiency and profitability in other domains, healthcare is relying on results-driven measurement (RDM) to drive reform. RDM utilizes execution methodologies (e.g. SixSigma, LEAN, TQM) that have revolutionized manufacturing and back office processes. However, when applied to healthcare, RDM suffers from fundamental data quality issues as a consequence of failing to apply the key enterprise architectural principle of alignment. RDM is provider-centric where desired results are determined by providers according to their individual, professional and/or consensus judgments. Evidence-based medicine (EBM), comparative effectiveness research (CER) and simple protocols are powerful methodologies for standardizing care but without alignment they can optimize population means to the detriment of population members. Healthcare needs to look beyond results to measure and deliver value from the functional, emotional, and spiritual perspective of the patient. Value-driven measurement (VDM) places the patient at the center by identifying and analyzing the jobs-to-be-done for which the patient is hiring their provider(s). VDM is holistic, cross-functional, and focused on optimizing measures for population members in order to enhance population means. Customer-driven healthcare can only be realized if value is measured from the perspective of the patient. While results are important, they are only optimal if they are aligned and derived from value measurements. Healthcare is being disrupted, giving way to more distributed models of service. Without VDM to align these changes across the continuum-of-care, reform efforts may increase the likelihood that patients' lives will be negatively impacted more than healthcare outcomes positively improved.

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1. HEALTHCARE REFORM AND PERFORMANCE-MANAGEMENT IN THE UNITED STATES.

It is often stated as axiomatic that you cannot manage what you cannot measure, and by extension that you cannot reliably and demonstrably improve that which you cannot accurately and quantitatively measure. These maxims were first taken to heart in manufacturing industries and they have subsequently demonstrated their worth in nearly every vertical within the business world. Reflecting these origins, improvement is largely understood in terms of the consistency of the processes that generate quality in a finished product. Different authors and schools of thought have sought to define this notion of quality and it is beyond the scope of this paper to review these or to provide clarity on their relative merits. It is clear that however process improvement and quality are defined, the concept of performance-management has revolutionized the practice of management science and, in its implementation performance management has dramatically improved the efficiency and profitability of business.

Healthcare as it is currently practiced in the United States is unsustainable (Chernew, Hirth, & Cutler, 2009), with costs rising faster than both GDP and per capita income for the last 50 years (Kaiser Family Foundation, 2009). Furthermore, for all its expense, advanced technology and cutting-edge techniques, the quality delivered by the U.S. healthcare system, whether assessed by measures of national health or by consistency of outcomes across individuals, is demonstrably lacking by comparison to other industrialized nations (Spithoven, 2009; Starfield, 2000). As such, the practice of healthcare in the U.S. is undergoing a tidal shift from its 19th century cottage industry model (Starr, 1984) of tradition- and ritual-bound care delivered by isolated and largely autonomous individuals (Adair et al., 2006; Beaulieu et al., 1999; Zitella et al., 2006). The final form of 21st century healthcare in the United States has yet to be determined, but it is certain that whatever this new form of healthcare, it will rely heavily on metrics which track cost-containment and quality-of-care and it will use the language and methods of performance management.

We will refer to this tidal shift, as the process of healthcare reform. We use this term in a non-partisan manner which makes no attempt to recommend or support a particular policy agenda or ideological viewpoint. This paper is solely concerned with how we can improve the measurement of value across the continuum-of-care to facilitate a balance between cost, efficiency and service which will transcend any particular change in healthcare resulting from either new technology or new policy.

1.1. BALANCING THE BUSINESS WITH THE PRACTICE OF HEALTHCARE

In recent years, the healthcare system has turned to performance management to provide accountability and drive improvement. Performance measures are designed to provide feedback to providers about their performance as compared with their peers and with established standards of care in order to drive improvement in the quality of healthcare decisions and outcomes. The use of Process-Driven Measurement (PDM) and Results-Driven Measurement (RDM) approaches in performance management has brought higher levels of care consistency, has reduced costs, and has provided payers with benchmarks from which to base care reimbursement. The benefits of PDM and RDM are very clear for the business of healthcare, however there have been concerns raised about their benefits in the practice of healthcare, especially as that practice relates to the health value delivered to a patient when quality is determined and managed solely by internal quality

improvement programs (Lohr, 1997). In general, PDM and RDM indicators are developed amongst providers, policy-makers, and payers. The patient is almost never involved in the selection of these indicators, in part because of the technical complexity of the indicators and in part because the indicators are designed in order for the provider to ensure the consistency of *their* processes and/or results. However, there is a growing recognition amongst providers and patients that healthcare needs to find a more patient-centered and therefore more holistic approach (Berwick, 2009a).

The purpose of this paper is to introduce a patient-centered measurement system which provides a method for identifying and evaluating the relative value of a patient’s various needs. This measurement system uses a methodical approach to define baseline value metrics from the patient’s perspective. The approach is captured in a well-defined model of the jobs-to-be-done that brings a holistic framework for, and a logical decomposition of, the healthcare domain. We believe this framework will restore a balance between the business and the practice of healthcare across the continuum-of-care for patients. Further, we assert that this approach, which we call Value-Driven Measurement (VDM), can be leveraged to drive improvement throughout the domain of healthcare and across the value network.

2. A GENERIC HYPOTHETICO-DEDUCTIVE APPROACH TO HEALTHCARE

All approaches to performance management have at their heart the hypothetico-deductive (H-D) method. In its most familiar form, this method proposes a four step algorithm (Figure 1.) for assessing whether a hypothesis is falsified by experimental observation (Godfrey-Smith, 2003). These four steps are: to gather observations from a baseline set of events; to formulate a hypothesis which explains these observations; to formulate and perform an experimental test capable of falsifying the hypothesis; and to observe the set of test events such that if the explanatory hypothesis is falsified, a new explanatory hypothesis should be generated, else formulate and perform a new experimental test capable of falsifying the hypothesis.

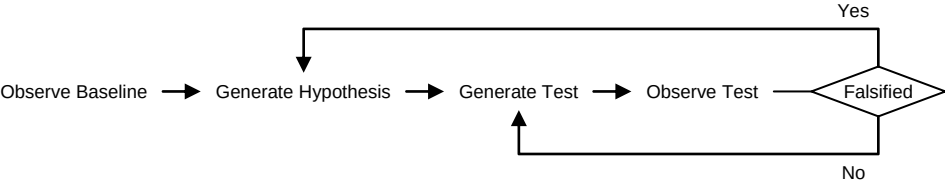


Fig. 1.The Hypothetico-Deductive Algorithm

Formal critiques of the H-D method have been undertaken by Kuhn (Kuhn, 1962) and Feyerabend (Feyerabend, 1975). These criticisms largely rest on the fact that no hypothesis exists in isolation from other hypotheses. While we acknowledge this to be implicitly true, we also believe that judicious use of the H-D method can be very productive for the generation of new, reliable and useful knowledge.

2.1. A SIMPLE MODEL OF AN EPISODE-OF-CARE

In order to extend the H-D method to healthcare, a simple model of the episode-of-care needs to be introduced. An episode-of-care may be defined as an “interval of care by a health care facility or provider for a specific medical problem or condition. It may be continuous or it may consist of a series of intervals marked by one or more brief separations from care” (Mondofacto Medical Dictionary, 1998). Figure 2. shows such a

simple model for an episode-of-care. This model consists of three components, *Presentation*, *Process* and *Outcome*.

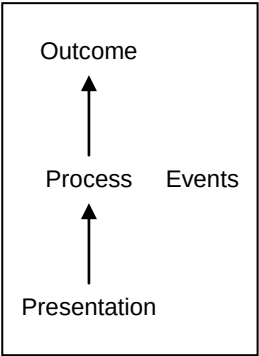


Fig. 2.A Simple Model of the Episode-of-Care

Presentation is the initial step in the episode-of-care, it is when a patient presents themselves, or is presented for care. This care may be anything from a routine check-up to an emergency room visit, to a phone call or an email requesting information or advice. Key features of *Presentation* are that it absolutely must precede the delivery of any care by a provider and that a patient must be involved. *Presentation* is a snapshot in time and it is the event which initiates the episode-of-care. *Process*, on the other hand, represents not just a single process but the entire set of processes that occur after *Presentation* and before *Outcome*. Processes may be performed by providers, by patients, by automated processes or by proxies. *Outcome* closes the episode-of-care in the same manner that *Presentation* begins the episode-of-care. Like *Presentation*, *Outcome* is a snapshot in time and it is the event which terminates the episode-of-care.

In this model *Presentation*, *Process* and *Outcome* are all events, that is they are real-world phenomena which exist outside of their measurement or observation. Furthermore, *Presentation* and *Outcome* may be thought of as initial- and end-state events, respectively. A common feature of initial- and end-state events is that neither can be directly modified, they are statements of conditions at two defined points in time. However, *Presentation* and *Outcome* are connected by a temporal flow. *Process* occurs during this temporal flow with the presumption that *Process*, in conjunction with *Presentation* determines *Outcome*. It is worth noting that individual initial- and end-states are ultimately dependent upon the essentially innumerable interactions between detailed and entirely unique systemic, molecular and perhaps even quantum mechanical states. For this reason, parametrically similar *Presentations* and *Outcomes* may be grouped into populations across multiple episodes-of-care and these populations may be understood in terms of their central tendencies and the variances around those tendencies.

It is worth emphasizing that in this model, each episode-of-care is a true system and displays all of the characteristics associated with the classical definition of a system: these characteristics are a boundary, an internal structure, behavior which processes inputs into outputs, and internal connectivity (Backlund, 2000). As with all systems, the definition of the boundary is somewhat arbitrary and can be drawn and re-drawn in many different ways. Rather than weakening the validity of the model, this strengthens its

utility through flexibility and explicit recognition of the inherent “systemness” of healthcare delivery (Berwick, 1992).

2.2. APPLYING THE HYPOTHETICO-DEDUCTIVE APPROACH ACROSS EPISODES-OF-CARE

It follows from Figure 2. that the only way, in our model of an episode-of-care, to effect the *Outcome* for a given *Presentation* is to change the intervening *Process*. Changes to one or more processes are made in order to promote a beneficial or desired *Outcome* according to what we are defining as a Hypothetico-Deductive Approach (H-DA). All H-DAs require at least two episodes-of-care, and likely many, many more as most formal implementations of the H-DA are based on some sort of statistical reasoning which incorporates notions of populations and/or sample size. The H-DA is heavily influenced by the work of Karl Popper (Popper, 1959).

Figure 3. graphically demonstrates how the H-D method and the simple model of episodes-of-care are combined to produce the H-DA. In this approach a set of baseline events are observed and from these observations a hypothesis is generated. The hypothesis leads to the generation of a test. As the test is performed a set of test events are observed and from these observations the hypothesis is assessed for falsification.

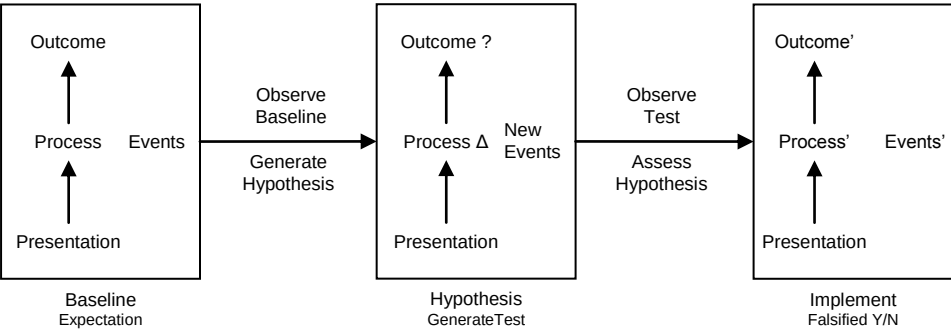


Fig. 3.The Hypothetico-Deductive Approach (H-DA) Across Episodes-of-Care

In Figure 3. there is an apparent parallelism between the Observe Baseline/Generate Hypothesis (the Observe/Generate) step and the Observe Test/Assess Hypothesis (the Observe/Assess) step. A more granular view of these steps reveals that both the Observe/Generate step (Figure 4a) and the Observe/Assess step (Figure 4b), do indeed share the following components: a Measurement System, an Analysis System, and an Execution System.

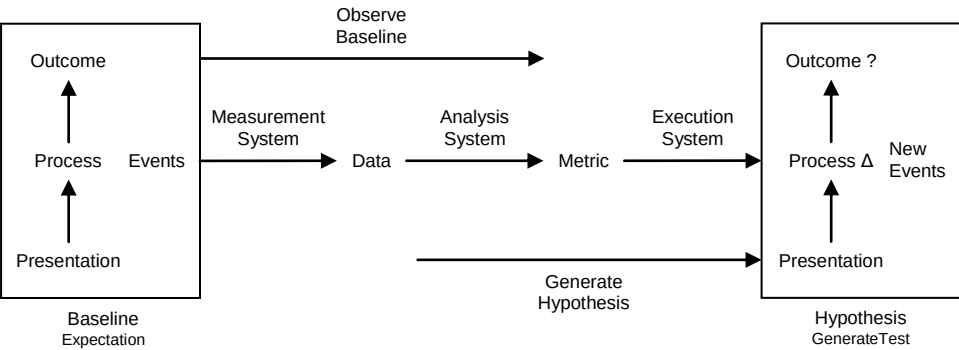


Fig. 4a.Granular View of the Observe/Generate Step in the H-DA

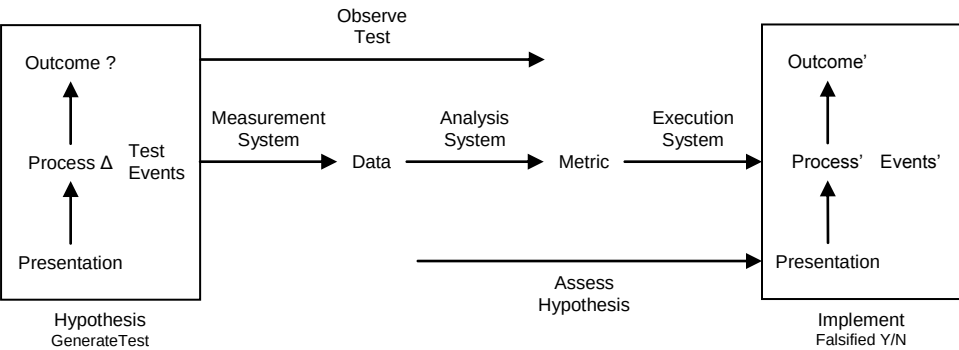


Fig. 4b.Granular View of the Observe/Assess Step in the H-DA

A Measurement System is the implementation of a specific Measurement Methodology, which methodology has a design based on specific assumptions (see Figure 5.). In this context, we understand data to be either qualitative or quantitative values captured by a Measurement System according to a specific Measurement Methodology and which data values, as part of that Measurement System, are either presented or stored in a particular architecture, which values and architecture may be accessed by one or more other systems. Simply put, the assumptions, design and implementation of the Measurement System determine the content, structure and context of the data captured.

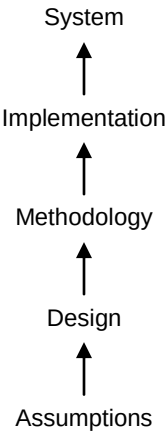


Fig. 5.Components of a System.

Like a Measurement System, the Analysis System is composed of assumptions and a design which comprise the Analysis Methodology and a specific implementation of that methodology. The Analysis System takes data from a Measurement System as its input and converts these data into one or more metrics. A metric then is data upon which a calculation has been performed; aggregation and arithmetic are common calculations although identity (i.e. a datum itself) and the application of procedural, conditional, statistical or bayesian logic are also all examples of calculations. As with data values, metric values may be either qualitative or quantitative. Metrics also have content, structure and context determined by the assumptions, structure and context of the Analysis System (n.b. the assumptions underpinning an Analysis System must include reference to the content, structure and analysis of the input data).

Together, the Measurement and Analysis Systems comprise the observation components of the H-DA, in Figure 4a we see that they comprise the Observe Baseline step and in Figure 4b we see that they comprise the Observe Test step. This makes sense as both observation components are capturing data and calculating metrics from a set of events.

An Execution System follows the same construction as Measurement and Analysis Systems, that it is composed of assumptions and a design which define the Execution Methodology and then a specific implementation of that methodology completes the system. An Execution System takes metrics as its inputs and executes some real-world change, for the Observe/Generate step, the real-world change is the performance of a set of test episodes-of-care; for the Observe/Assess step, the real-world change is either the creation of a new hypothesis (if the current hypothesis is falsified) or the creation and performance of a new set of test episodes-of-care (if the current hypothesis is not falsified). Where the H-DA differs from Figure 1. is if the *Process A* has yielded an acceptably beneficial improvement over the expected *Outcome*, then the *Process A* becomes the new process, *Process'*, with a new expected outcome, *Outcome'*.

In this model, the Analysis System is shared by both the Observe and Generate steps and then also by the Observe and Assess steps. This reflects the fact that the observation of events and hypotheses about those events are interrelated phenomena. In this sense both Kuhn (Kuhn, 1962) and Feyerabend (Feyerabend, 1975) were correct that no

observation can be made without a pre-existing framework of hypotheses and that to some extent the act of observation in the hypothetico-deductive method of Figure 1. presupposes the particular hypothesis to be tested.

H-DAs often weave together Measurement, Analysis and Execution Systems, as well as Baseline and Observational Episodes-of-Care in web-like networks, especially when less systematic and less formalized H-DAs are utilized. H-DAs are usually categorized along two axes either according to their Execution Methodologies and/or according to their measurement drivers. Some common Execution Methodologies are SixSigma, LEAN, Total Quality Management (TQM), care templates, protocol checklists, peer-review, evidence-based medicine (EBM), comparative effectiveness research (CER) and even traditional clinical intuition. The most common drivers are process, results and value. In practice, these Execution Methodologies and drivers are rarely so cleanly separated and more often are mixed-and-matched according to expediency and expertise.

Given that process improvement is a journey with no end, and that both false-positive and false-negative results are possible, it is useful to introduce the concept of a view. A view comprises some bounded set of one or more of each of the H-DA elements discussed above.

Another way to think of views is in terms of the actual transactional, operational and analytical systems which capture and process data and present metrics. These systems are rarely implemented as cleanly and compartmentally as described above. A single transactional system may perform measurement, analysis and execution. In this context then, the view is defined by the set of transactional, operational and/or analytical systems. It is worth noting then that the same value may be both a datum and a metric depending on the depth of lineal analysis encompassed by a particular view (Lee, Pipino, Funk, & Wang, 2006). Another way of understanding the distinction between a data value and a metric value is to ask whether the current view contains enough context to unambiguously decompose a particular value into two or more constituent parts. If a value cannot be decomposed, it is likely to be a datum in the view, whereas if the value can be decomposed, it is likely to be a metric. In general, the decomposed parts are generated and/or held as distinct data and/or metric values somewhere else in the view.

2.3. SOME ASPECTS OF DATA QUALITY: FITNESS IN THE H-DA MODEL OF HEALTHCARE

Many authors have sought to define, classify and quantify the notion of quality as applied to either data or information (Madnick, Wang, Lee, & Zhu, 2009). While these definitions and schemata differ in many respects they all agree that quality is a multi-dimensional entity, identifying anywhere from 4 to 26 or more distinct dimensions of data quality (Redman, 2004; Wand & Wang, 1996).

Fitness is a key concept which appears in both the academic and industry data quality literatures; and it will be the focus of the remainder of this paper. Juran has defined "...data to be of high quality if they are fit for their intended uses in operations, decision making and planning" (Juran & Godfrey, 1998). Redman has extended Juran's definition, stating that data that is fit-for-use is free of defects and possesses desired features (Redman, 2001). In both Juran's and Redman's assessment then, quality and *fitness-for-use* are synonymous terms. We therefore find fitness in the sense of fit-for-use to be an unproductive term that falls outside of fitness as we seek to discuss it here.

Fitness, as a productive term distinct from quality, comes in two flavors, *fitness-for-purpose*, sometimes also known as fit-to-purpose, and *fitness-of-purpose*. In general *fitness-for-purpose* is the degree to which a product or process meets its specifications,

whereas *fitness-of-purpose* is the degree to which these specifications support the production of a desired outcome. Such a definition begs the question of who determines the specifications and the desired outcome. Ideally the desired outcome should be set by the consumer of that outcome and the specifications should be set by an architect with the appropriate skill sets, technical knowledge and experience to bridge and balance people, process and technology across the organizational, cultural and functional silos of an enterprise. This definition is an extension of fit-for-purpose as used in the Department of Defense Architectural Framework 2.0 (Department of Defense, 2009).

From the perspective of a particular datum in the H-DA, *fitness-for-purpose* captures the adherence of that datum to the specifications defined in the design of the Measurement Methodology. Note that less than perfect *fitness-for-purpose* of a given datum results from one or both of two causes: the first being that the implementation of the Measurement System does not match the design of the Measurement Methodology; the second being that the event being captured by the Measurement System violates one or more assumptions implicit in the design of the Measurement Methodology.

From the perspective of a particular datum, *fitness-of-purpose* captures the appropriateness of that datum for the calculation of a specific metric. The calculations and algorithms which operate on data to generate metrics contain their own assumptions, assumptions about the structure and context of the underlying data. Should the underlying data violate any of these assumptions, then the *fitness-of-purpose* of that data for the metric in question is poor.

From the perspective of a particular metric, *fitness-for-purpose* captures the suitability of that metric to the specifications defined in the design of the Execution Methodology. As with the *fitness-for-purpose* of a particular datum, the *fitness-for-purpose* of a particular metric is determined by either the alignment of design and implementation within the Execution System and/or by the adherence of the metric values to the assumptions that underpin the design of the Execution Methodology. Should the content, structure and/or context of a metric violate one or more of these assumptions then it is not *fit-for-purpose* for that Execution System.

Finally, there is the *fitness-of-purpose* for a particular metric. The selection of which metrics to track in a H-DA is determined by three things: 1) the baseline expectation and hypothesis; 2) the Execution System; and 3) any physical limitations or resource constraints that would prevent events of interest from being measured or analyzed and/or a *Process Δ* from being implemented. *Fitness-for-purpose* of the metric encapsulates each of these determinants (baseline, execution and limitations), *fitness-of-purpose* of the metric instead asks what relation does the metric, and any associated *Process Δ* , have to the *Outcome* and what is the value of that *Outcome* to the consumer. Oddly enough, in some sense, the *fitness-of-purpose* of a metric should be the first fitness assessment which is made, especially in healthcare settings. That it is rarely, if ever, assessed begs the question why and to understand this why, we must discuss Process-Driven Measurement (PDM), Results-Driven Measurement (RDM) and ultimately Value-Driven Measurement (VDM).

3. PROCESS-DRIVEN MEASUREMENT (PDM) IN THE H-DA MODEL OF HEALTHCARE

Process-Driven Measurement (PDM) is the oldest branch of performance management and still most closely reflects its origins in traditional manufacturing. PDM begins with Taylor's principles of scientific management (Taylor, 1912), Henry Ford's assembly line innovations (Rae, 1969), Walter Shewhart's mathematics of statistical quality control

(Shewhart, 1939) and W. Edward Deming's studies of quality and productivity (Deming, 1988). Under Deming's tutelage, Taiichi Ohno, Shigeo Shingo, and Eiji Toyoda, brought the principles of Shewhart and Deming to their first full fruition in the Just-in-Time (Shingo, ShingÅ •, & Dillon, 1989) and then in the Toyota Production System and LEAN methodologies (Ohno, 1988).

For the purposes of the H-DA model of healthcare, the distinguishing feature of PDM is its focus on measuring and implementing process changes in real- or near-real-time. Individual process workers, those whose intimate knowledge of the process comes from actually performing the process, are empowered to identify waste, that is inefficiencies in the processes for which they are responsible, and to recommend solutions to reduce that waste. These solutions are rapidly implemented and their effect on waste reduction is quantified. The assumption underlying this approach is that reduced waste in the production process leads to increased quality in the final product and that increased quality of the final product leads to increased stakeholder satisfaction where the stakeholders are the process workers, the consumers of the product and ultimately the owners of the production company (shareholders in the case of a public company, citizens in the case of government, and owners in the case of private companies and non-profit entities).

3.1. DATA QUALITY IN PDM: FITNESS AND MEASURE ALIGNMENT

As previously discussed, fitness comes in two flavors, *fitness-for-purpose* and *fitness-of-purpose*. In the PDM context, *fitness-for-purpose* can be readily and very precisely managed. The robust methodologies and decades of real-world experience and implementation behind PDM ensure that the assumptions and design consideration behind Measurement, Analysis and Execution Methodologies are well-understood and display a high-degree of coherence. The many successes of PDM are well-documented and too numerous to cite. PDM has even been extensively implemented in the healthcare setting with dramatic results (Bentley, Effros, Palar, & Keeler, 2008; Weber, 2006).

However, pointed critiques of PDM have also been made. The most relevant being that PDM overemphasizes processes by asserting that the quality of the product, or the quality of the *Outcome* in the H-DA model, is solely dependent upon the minimization of waste in the performance of the constituent processes. It has been stated that there is no necessary connection between process consistency and outcome quality (Schaffer & Thomson, 1992). These authors argue strongly for the replacement of process with results (i.e. *Outcomes*) while other authors argue for an approach which requires less cultural change than is required by the adoption of traditional PDM (Barnsley, Lemieux-Charles, & Baker, 1996; Leggat et al., 1998).

Fundamentally this disconnect between process consistency and outcome quality arises from a misalignment between the process optimization methods and the outcome optimization methods. The proposed solution, to align process measures with outcome measures forms the basis of RDM and will be discussed in more detail in Section 4.

3.2. DATA QUALITY IN PDM: PROCESS MEASURES AS A BY-PRODUCT

In order for information to be managed as a product, four principles must be followed. The needs of the information consumer and the purpose to which the information will be put must be well understood. The process of creating the information must be managed as a production process. The lifecycle, from creation through maturity and decline, of the information must be defined and managed. And finally, a responsible party needs to be

identified as the owner of the information process and its products (Lee, Pipino, Funk, & Wang, 2006; Wang, 1998).

Again, the emphasis in PDM is on the processes which produce the product. Any data or metrics collected as part of PDM are solely in the service of identifying and reducing waste in one or more processes. This tends to lead to situations where the data and metrics are by-products of the production process rather than being treated as products in-and-of themselves. It has been shown that when information (data and metrics) are treated as by-products rather than products, data quality suffers (Wang, 1998).

As an indication of the degree to which the needs of the information consumer are ignored, Lied and Sheingold (Lied & Sheingold, 2001) noted in a well-received review that "... (t)here are real concerns that the act of measurement itself has taken on such a symbolic significance over and above the power of such information to promote beneficial and worthwhile change. We do not yet know how to make such systems deliver on the promises made for them."

3.3. DATA QUALITY IN PDM: SUBORDINATION OF THE PRODUCT TO THE PROCESS WORKER

A fundamental underpinning of Shewhart's statistical quality control is that the events being measured and analyzed are normal and unimodal, displaying, for any given metric, a sample mean, $\bar{X}$, which approximates the population mean, μ , and a sample variance, s , which approximates the population variance, σ (Shewhart, 1939). When a process change is made which affects quality, the new sample will either exhibit a change in $\bar{X}$ and/or a change in s for one or more process metrics. Those process changes which yield a statistically significant increase in quality as measured by improvements in $\bar{X}$ and/or s are deemed to be quality improvements. These process changes are then enshrined as the new best practice and all subsequent production work is done according to the new process.

At first blush making such changes seems to be eminently reasonable. And from the perspective of the process worker, these changes are. This approach will reduce waste in their process thereby increasing the process worker's efficiency, consistency and accuracy in the performance of their process. However, PDM makes no demand on the process worker to understand or to consider anything more, and absolutely nothing less, than the process for which they are responsible. Each process worker therefore rushes towards a local minima of waste for their process while potentially compromising one or more aspects of the overall production process. In short, the quality of the product (the outcome in our H-DA model) is subordinated to the optimization of the sample characteristics of the processes; the outcome is subordinated to the process worker.

Berwick's (Berwick, 2009b) automobile analogy is helpful here: "(a)n automobile that is designed to achieve a maximum speed of 100 miles per hour will not fare well in a NASCAR race. This system - the car - is designed for a performance level that is inadequate for this purpose. Its top speed is a predictable property of the system, and no matter how hard we push this system, it cannot possibly achieve the desired performance given its current design. Going faster requires a different system ... (, i)mprove the design to change the result." Now imagine a crew of mechanics descending on the car, each with their own specialty and each intent on optimizing one element, whether it be the transmission, the suspension, the engine, the exhaust, the body and so on. But they must work one at a time and they are only allowed to talk first to the mechanic who precedes them, and then, after they are done with their modification, to the mechanic who will follow. The car that comes out at the end of the process may go 160 miles per hour,

it may go 60 miles per hour, it may go 200 miles per hour for 10 minutes, or it may not go at all.

Furthermore, both $\bar{x}$ and s are very sensitive to distortion by the effects of outliers, non-homogenous or incomplete sampling across study blocks, multimodality, and deviations from normality such as skewness, kurtosis and heteroskedasticity (Daniel, 1999). Health care processes and in particular the challenges to, and responses of, the human body are notorious for generating each of these effects, both in the atomic (the individual patient) and in the aggregate (the patient population) - as many a provider readily points out, patients are not widgets. Without careful experimental design and appropriate statistical rigor it is very easy to violate the assumptions of either the Measurement, Analysis and/or Execution methodologies (Ibrahim, 2001; Zaslavsky, 2001) and in so doing to invalidate, or at least cast serious doubt, upon any conclusions regarding the quality implications of a particular process change.

Subordination of the product to the process worker illustrates how misalignment and the failure to treat process data and metrics as products conspire to produce not only poor-quality data, but also poor quality H-DAs and poor quality outcomes.

4. RESULTS-DRIVEN MEASUREMENT (RDM) IN THE H-DA MODEL OF HEALTHCARE

Results-Driven Measurement (RDM) is a direct response to the limitations of PDM. RDM moves the focus of measurement, analysis and execution from processes to outcomes. *Process* is still vitally important to RDM as changes in processes are still the only way to affect changes in an expected *Outcome*. However, in RDM there is less emphasis on cultural change and more emphasis on utilizing the "...abundance of underexploited capability and dissipated resources..." that exists in most organizations (Schaffer & Thomson, 1992). RDM also begins to approach performance management as the management of a system, in this case, as the system which produces the currently expected outcome. The first step in RDM is always the selection of one or more candidate outcomes of interest.

The practice of modern medicine is overflowing with events, data and metrics, whether captured or calculated in paper or electronic charts, whether structured, unstructured or semi-structured, whether observed in inpatient, outpatient, laboratory or care management settings, whether canonized in textbooks, reported in journals or passed on through tradition and ritual (Beaulieu et al., 1999; Lomas et al., 1989; Zitella et al., 2006). The depth and breadth of input available to a provider making a care decision is overwhelming. In such a context any set of tools that brings an approach which is structured, repeatable, and able to produce reliable predictability will yield not only measurable, but likely dramatic, improvements in care, and in fact such improvements have repeatedly been shown for a variety of different RDM approaches, including everything from traditional case- and peer-review (Goldman, Demaso, & Kemler, 2009; McKay, Bradley, Lough, & Bowie, 2009; Neale, Woloshynowych, & Vincent, 2003), to simple checklists (Hales & Pronovost, 2006) to EBM (Berwick, 2005), to CER (Chalkidou et al., 2009; Drozda et al., 2009; Sox & Greenfield, 2009) and classic double-blind studies (Berwick, 2008; Veronesi et al., 1981).

4.1. DATA QUALITY IN RDM: FITNESS AND STAKEHOLDER ALIGNMENT

Even a cursory survey of the performance management literature for healthcare quickly reveals the preponderance of providers with deep medical and academic credentials. Given the complexity, specialization and power of modern healthcare, such a distribution

of authorship is to be expected and perhaps even desired. However, this technical emphasis creates a significant misalignment.

Alignment is a key enterprise architectural principle that is closely related to fitness. In fact, assessing *fitness-of-purpose*, especially in the context of metrics, requires an assessment of alignment. The discipline of Enterprise Architecture (EA) promotes enterprise-level assessment of contribution to, and rationalization of, strategic goals across all stakeholders within an enterprise. Stakeholder contributions are often at odds due to miscommunication and misunderstanding regarding the desired goals. Politics, personality and culture are also all causes of misalignment between stakeholder contributions and enterprise goals. Perfect alignment occurs when all stakeholders share a full and common understanding of the goals and collaboratively work together to achieve them. Perfect alignment is rare if not impossible to achieve, but the practice of EA has many well-established techniques to improve alignment (Sousa, Pereira, & Marques, 2005; Thomas et al., 2000).

Central then to the question of alignment is, who are the stakeholders? In an episode-of-care, the absolutely requisite stakeholders are the patient and the provider (where the notion of provider includes care-providing institutions as well as care-providing individuals). The traditionally paternalistic ethos of medicine (Tan, 2002) as well as the efficacy of provider-delivered processes has emphasized the centrality of the provider in the setting of goals. As discussed above this approach has been very effective in quantitatively improving care (Leape & Berwick, 2005) by focusing on the technical aspects of care delivery, process improvement and outcome optimization. Notably missing though is any explicit involvement of the patient as a stakeholder.

The usual presumption of the provider is that a beneficial outcome, any beneficial outcome is of value to the patient. Furthermore, as providers are trained to think in the terminology and models of their areas of expertise, they tend to formulate their goals in these same terminologies, utilizing highly specialized models to generate hypotheses. Most patients are neither trained nor familiar with these terminologies and models and so they can easily formulate goals that might appear irrational or valueless to the provider but which the patient values highly (Berwick & Weinstein, 1985). Intriguingly, there are even situations where attention given by the provider to the spiritual and emotional needs of the patient can have a dramatic effect on patient outcomes (Egbert, Battit, Welch, & Bartlett, 1964).

Another source of misalignment is the inclusion of payer goals, especially where the payer is neither the patient nor the provider. When the payers are third-party insurers, they tend to use the same terminology and models as the providers, however whereas the providers' emphasis tends to be on processes which will maximize the likelihood of beneficial outcomes while simultaneously minimizing the likelihood of harmful outcomes, payers' emphasis tends to be on maximizing the likelihood of beneficial outcomes while minimizing the costs associated with achieving that outcome. A fuller treatment of the misalignments arising from the inclusion of payer goals is beyond the scope of this paper.

4.2. DATA QUALITY IN RDM: OUTCOME MEASURES AS BY-PRODUCT

RDM's focus on measuring and managing outcomes allows for more room to treat information as a product. However, the RDM approach emphasizes the selection of particular result measures according to a provider and/or payer-centric viewpoint. This bias tends to emphasize isolated, and highly technical, components rather the holistic outcome. One of the key criteria for managing information as a product is to "(treat)

them as an integrated whole (from) a cross-functional perspective” (Lee, Pipino, Funk, & Wang, 2006). Ignoring or inadequately considering the perspective and needs of the patient, by definition ignores the whole – failing to collect and analyze data relative to the patient’s jobs-to-be-done is to deliberately ignore the healthcare consumer, as well as a key healthcare producer. In addition to analyzing how the data must be managed, the Information Product approach requires appropriate answers to what data is managed, why is this data managed, what determines successful management and finally who manages this data (Lee, Pipino, Funk, & Wang, 2006). The answers to these five questions are factors as central to the management of information as a product as is adherence to the four principles discussed in Section 3.2.

4.3. DATA QUALITY IN RDM: SUBORDINATION OF THE PATIENT TO THE EXPERT

RDM corrects for PDM’s overemphasis on the process worker, and it eliminates one of the key criticisms of PDM – that there is no necessary causal relationship between statistical quality control of individual processes and the final product produced (Schaffer & Thomson, 1992). With RDM, by definition, a change in the process which produces a change in the product is a causal effect (within the bounds of statistical inference and uncertainty). RDM is somewhat more difficult to implement however as each RDM effort must begin with a guess, namely that one or a handful of process changes will have, in fact, a statistically significant causal effect on the outcome. PDM’s great strength is that it does not require any such presuppositions.

Returning to Berwick’s automobile, whereas PDM argued that simply optimizing each individual component of the car would lead to high performance, RDM requires that some subset of components, and specific modifications to those components, be selected *a priori*. Given a reasonable understanding of the mechanics of the automobile, the state of the art in automobile engineering and some good old-fashioned intuition and hard work, the original car can be redesigned and modified to perform to just about any criteria which violate neither the laws of physics nor the limits of human psychobiology. For all the apparent sensibleness of this approach and despite its appeal to our preference for technologically engineered solutions, there are three fundamental problems. First and foremost, the car’s owner was never asked for what jobs they were hiring their car, there is after all little use in converting a minivan or a pickup truck into a NASCAR racer. Second, the physiological and cultural characteristics of the particular driver were never taken into account, the six-foot-eight, British stick shift driver will require a very different vehicle than the four-foot-eleven American automatic driver to achieve the same level of high performance. Third, the limits of high performance will differ by owner, there will always be some drivers for whom a high performing automobile will pose hazardous either to their own health or to the health of others and/or other drivers for whom even when presented with the highest performance automobile will still never drive it faster than 15 or 20 miles per hour.

So, just who is selecting the components and designing their modification? As discussed in Section 4.1, it is the providers and the payers, utilizing highly technical, very sophisticated and extremely domain specific models. What the payer- and/or provider-centric goals of RDM have in common is an emphasis on the optimization of particular metrics. Quality of care is then determined by both relative and absolute performance on process, and ultimately on outcome, metrics. Process improvement is axiomatically defined as an improvement in one or more of these metrics. In short, improvements in

the metrics are ends in and of themselves, with the means of achieving these improvements subordinated to the up- and down-ticks of these ends.

To the degree that RDM also makes use of the same statistical quality control methods as PDM, RDM is susceptible to issues of outliers, non-homogenous or incomplete sampling across study blocks, multimodality, and deviations from normality such as skewness, kurtosis and heteroskedasticity (see 3.3 for details). Due to its emphasis on improvement of outcome over process, RDM is also susceptible to a relatively recently identified statistical anomaly known as the “Will Rogers Phenomenon” (Feinstein, Sosin, & Wells, 1984). The American humorist Will Rogers once quipped “When the Okies left Oklahoma and moved to California, they raised the average intelligence in both states” (Christensen, 2003). Clinically, the Will Rogers Phenomenon most often occurs between classification schemes with differing inclusion criteria and/or differing diagnostic sensitivity. The most important thing to note about the phenomenon is that it is entirely artifactual – in Will Rogers’ example, the average intelligence across both states does not change, even though the state-specific averages do. Despite the artifactual nature of the Will Rogers Phenomenon its effect is not negligible, at least one study has identified a shifting comparable to the observed difference between current multiple sclerosis therapies and placebo (Sormani et al., 2008).

It is the slavish focus on improvements in outcome metrics which leaves RDM open to this artifact. To the extent that outcome metrics represent the full expression of a provider or payer’s model of care and their most valued result, RDM subordinates the patient to the expert.

5. VALUE-DRIVEN MEASUREMENT (VDM) IN THE H-DA MODEL OF HEALTHCARE

Value Driven Measurement (VDM) changes the approach to measurement from siloed to holistic. RDM attempts to define what is best for the patient by focusing on the functional outcomes at a macro level. While these are certainly critical for achieving success from the perspective of the provider and payer, as discussed in Section 4, they often fall short of addressing needs from the patient’s perspective. This is where VDM fills the void by defining the desired outcomes that patients want to achieve and then assessing how well the services delivered address the patient’s desired outcomes. VDM creates a complete picture of the experience that the patient wants to achieve within the scope of a job that the patient is trying to have done. Patients expect the provider to accurately deliver the functional jobs for which the provider was hired to perform. However, this is not enough. Patients also need and expect their emotional and spiritual needs to be addressed as well. This is evident from case histories where highly skilled physicians are sued by their patients simply because the provider did not listen to their needs (Wojcieszak, Banja, & Houk, 2006). In other cases, the actual patient outcomes from a procedure may have been sub-standard, yet the patient refuses to sue their physician because the physician has established a “bond” by attending closely to the patient’s emotional and spiritual needs (Cantor, 2002; Edwin, 2009; Kaldjian et al., 2007; Levinson & Gallagher, 2007).

Once we can create a complete picture of the patient’s desired outcomes for a given job we can layer in RDM and PDM to align the clinical and institutional perspectives with those of the patient. This alignment of perspective drives alignment of data and alignment of service delivery amongst providers across the continuum-of-care.

A relevant example of how this alignment can benefit healthcare is being played out today. The HITECH provisions of the American Recovery and Reinvestment Act

(ARRA) (111th Congress, 2009) have caused a torrent of investment into electronic medical records (EMRs) and health informatics (HI), but most analysts agree that the value for patients will not be realized simply through the implementation of an EMR, even one that complies with the evolving definition of Meaningful Use (MU) (Hoehn, 2010; Kadry, Sanderson, & Macario, 2010; Porter, 2009). The key selling points for these technologies are the improvements in the quality and quantity of clinically-relevant data. The methods of data capture will continue to improve as technology continues to be infused into the delivery of healthcare services. Eventually, the integration of technology and information will begin to improve the delivery of healthcare services. We propose that leveraging VDM can help to expedite the realization of value for providers and their patients. The insights gained from VDM can be used to align process-driven and results-driven metrics for both providers and patients by prioritizing the investments in EMR/HI that address the important (i.e. high-value) outcomes. The features that address these important outcomes first are the ones that should be prioritized for investment and improvement initiatives. This will ensure a patient-centric view is maintained throughout the development cycles for EMR and HI. Leading with VDM will increase the likelihood that investments in both EMR and HI will address market opportunities (resulting in greater value for the entire value network) and will help to optimize the opportunity pipeline, ensuring that healthcare dollars are well spent.

VDM is based on a perspective of measuring the desired outcomes that an individual would like to achieve within the context of a particular "job" that he/she is trying to get done. Work is typically classified by roles, titles, and policies (articulated as standard operating procedures, practiced guidelines, and/or best practices). These classifications are designed to compartmentalize the work that needs to be done by placing artificial boundaries (which may or may not be logical) and to ensure separation of responsibility in order to support accountability and build a taxonomy of jobs in the healthcare domain model (Atkinson & Abu El Haj, 1996).

What people are actually trying to "get done" is often a combination of various procedures, complex interactions, and discrete activities designed to help prepare, execute, verify, and complete a given job (Christensen, Anthony, Berstell, & Nitterhouse, 2007). These job-related activities include a variety of discrete tasks (many of which are common amongst other jobs) which are used by every job executor to plan, prepare, execute, verify/monitor/adjust, and complete a given job. These common tasks can be abstracted in a reference model for modeling a job (Lewis, 1998).

The reason for introducing the term "job" is that it allows us the ability to abstract the focus of work from the myriad of complex procedures and policies to a set of commonly applied steps that allow an individual to achieve one or more desired outcomes. The specific tasks performed in any one given job may vary from person to person based on *who* is actually executing the job. However, the job itself becomes a universal constant in terms of achieving the desired outcomes.

This is a very powerful concept and allows the application of architecture to build structure, reuse, alignment and simplification of very complex relationships and seemingly disparate and/or unrelated activities. Once we can impart structure, we can begin to formulate patterns-of-work to explain, anticipate, and measure the jobs that individuals are trying to get done within a given area, location, and even domain. The work of an Enterprise Architect (EA) in this arena is not altogether too dissimilar from that of a physicist when trying to explain the seemingly unstructured phenomenon of activity within the domain of space (Gentner & Stevens, 1983).

By establishing the "job" as the primitive object, we can define an ontology for healthcare across all the sub-domains (prevention, diagnosis, treatment, maintenance) which ontology will provide a consistent model regardless of how healthcare evolves over time. This is vitally important to measure value across the continuum-of-care as healthcare is rapidly being disrupted by both technology and new business models (Christensen, Grossman, & Hwang, 2008).

5.1. DATA QUALITY IN VDM: FITNESS AND HOLISTIC ALIGNMENT

RDM and PDM provide data for use in monitoring and assuring healthcare consistency, process compliance, and results that are within acceptable tolerances. These are all valuable and necessary metrics and objectives. VDM can extend this value by bringing a holistic measure of patient data that can be used to drive improvement and innovation in healthcare by aligning the RDM and PDM metrics to ensure they address value as defined by the patient. RDM and PDM are founded on the principle of *fitness-for-purpose* thereby assuring that quality is met by process and/or procedure if the data demonstrates alignment with a specific provider-centric outcome (Vlăsceanu, Grünberg, & Pârlea, 2007). VDM is focused on Lester's definition of *fitness-of-purpose* (Lester, 1999), which is congruent with our definition from Section 2.3, with the aim of defining clear measurements of value from the perspective of the patient. The goal of VDM is to remove bias and ambiguity in determining the important patient outcomes, thus raising the quality of *fitness-of-purpose* to much higher levels. By modeling this data within a well defined ontology and incorporating good practices of data architecture, VDM can be positioned to guide the alignment of data silos across the healthcare continuum and provide a baseline for aligning services throughout healthcare (thus providing both vertical and horizontal integration). Once we can achieve alignment of data, we can begin to align the services provided by providers. This will lead to an organic model of healthcare reform that is patient-focused. Incremental improvements and innovation (including disruptive innovations) can proceed with the goal of *fitness-of-purpose* to ensure desired outcomes for patients are used to direct the *fitness-for-purpose* efforts by quality agencies.

5.2. DATA QUALITY IN VDM: VALUE MEASURES AS PRODUCT

The Information Product approach requires that data be managed according to the four principles discussed in Section 3.2 and furthermore that the five factors discussed in section 4.2 must be analyzed. Whereas PDM violates essentially all four principles and RDM incorrectly emphasizes management of component parts, VDM meets all four principles and correctly addresses the five factors.

The first of the four principles is to understand the information consumer's needs; VDM explicitly addresses the needs of all stakeholders. The second principle is to manage information as the product of a well-defined information production process; by emphasizing appropriate alignment across all stakeholders VDM produces consistent, coherent information products. The third principle is to manage the life cycle of the information product; by continuously focusing on the ultimate consumer of healthcare and the dynamic nature of value, VDM adapts metrics and measures to evolving needs and technologies. And finally, the fourth principle is to appoint an information product manager to manage the information production process; the governance models required for effective VDM require that appropriate individuals be held responsible across the entire information production process.

In terms of the five factors: VDM manages information across its lifecycle; VDM manages the information product in an integrated, cross-functional and holistic manner;

VDM manages the information product to deliver high-quality to information consumers; VDM defines success over the continuum-of-care and therefore delivers continuous quality over the information product's lifecycle; and finally dedicated and responsible parties manage the VDM information product over time.

5.3. DATA QUALITY IN VDM: BALANCING THE PROCESS, THE PROVIDER AND THE PATIENT

VDM provides healthcare with a transitional strategy to migrate care from the process worker and provider-led approaches of PDM and RDM to a more effective patient-led approach. Here the desired outcomes for the job of care move from the process-worker and/or provider to that of the patient. Because the executors (i.e. process-worker, provider and patient) are different, the desired outcomes will be different. The patient will only be able to make a good decision once the desired outcomes that they are trying to achieve as they execute this "new" job are well understood. To understand and capture the desired outcomes and align the solutions a VDM study can be performed to ensure that the patient is best positioned to perform the job. Looking towards the future, the focus can be expanded beyond patient outcomes in order to begin to assess provider outcomes, ensuring the delivery of the right services, at the right time to address patient outcomes. VDM is both reciprocal and recursive by nature, which further extends the value of the measures once they are warehoused and properly attributed. The patterns that can be extracted to promote continuous improvement and drive innovation can provide a valuable source of intellectual property over time.

If PDM begins by tasking each individual mechanic to optimize their component of Berwick's automobile, and RDM begins with an engineer planning to modify specific components of Berwick's automobile, then VDM begins with a detailed assessment of the capabilities, needs and jobs-to-be-done of the owner and driver of the automobile. Once the job-to-be-done is well understood then the techniques of RDM and PDM can be applied to modify and optimize the components of the automobile. In this manner the suburban family will get their minivan, the rural construction worker will get their pickup truck and the professional racer will get their stock car. The six-foot-eight English driver will get extra legroom, a lower seat, an airbag with greater compression resistance, a steering column on the right-hand side of the car and a left-handed stick shift while the four-foot-eleven American driver will get better access to the pedals, a lower dashboard for improved visibility, an airbag with lower deployment energy to prevent injury, a steering column on the left-hand side of the car and an automatic gear shift on the steering column. Notice how once the driver's (i.e. the patient's) needs are well-understood, the efforts of the engineers (i.e. providers) and the mechanics (process-workers) align naturally and self-evidently with each other and with those of the driver. By validating the *fitness-of-purpose*, *fitness-for-purpose* is optimized ensuring the highest-possible quality.

5.4. VDM METHODOLOGY

The VDM methodology is a 6-step progression of planning, execution, and analysis patterned after Strategyn's methodology (Ulwick, 2005), which is a proven methodology for product-based manufacturing organizations, including many healthcare provider institutions. We have extended the methodology to impart a consistent taxonomy in order to drive a holistic data architecture and to provide a foundation for data alignment and data mining to support pattern extraction using formal techniques in multi-

dimensional analysis. Figure 6 provides a schematic of the architectural decomposition of VDM in healthcare and illustrates its alignment with other measurement frameworks:

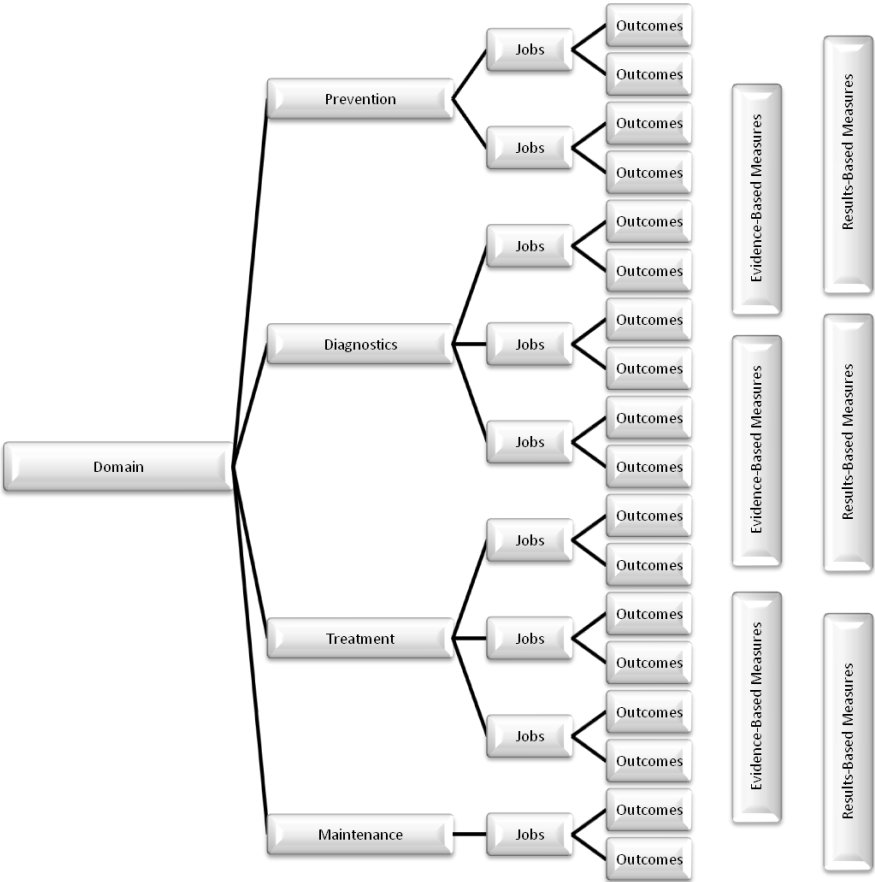


Fig. 6. Architectural Decomposition of VDM

The data captured using VDM will, over time, fill in the acknowledged theoretical (Baker, 1997) and practical (Panvelkar, Saini, & Armour, 2009) gaps in the understanding of what is value to a patient.

The following briefing provides a walkthrough of the VDM methodology:

5.4.1. *Definition Of The “Job”*. We start with the “Job” as our foundational “building block” from which to build a bottom-up model of the jobs within the healthcare domain. The ontology can be built gradually over time and will grow organically as institutions pool their studies together and outcomes are aligned across the various jobs within an individual organization’s value network (value networks can be conceptualized as a multi-dimensional supply chain).

A more formal effort for modeling these jobs will emerge as a natural outcome of driving data consistency. As in any data architecture, the reference architecture and core modeling provides some of the greatest opportunity and challenges (Frigg,

2010). The scope of this paper is to present the methodology and raise visibility on the need for a formal effort to model healthcare domains.

Adherence to data modeling and data architecture is a common thread throughout the VDM to ensure that artifacts (including, but not limited to products, services, processes, and information) can be mapped across the healthcare domain to support analysis and alignment with RDM and PDM.

Within scope of a VDM project, the focus is primarily on understanding the job (or jobs) that need to be studied. For example, if a hospital wanted to focus on improving their pharmacy operation they might conduct a study to first define the jobs that are performed by the pharmacists and pharmacy technicians. This leads into the next step, which outlines the scope for the study in greater detail.

5.4.2. Location. Step 2 attempts to locate the intersecting jobs across the spectrum of experience that can influence desired outcomes for the core functional jobs being studied.

Once the core functional jobs have been identified for the study, an analysis is performed to identify the other functional jobs that directly or indirectly impact the core functional job. Going back to the pharmacy example above, there are other functional jobs performed by nursing staff and physicians that may impact the outcomes for the job being studied in the Pharmacy operation of a hospital. There are also supply-chain oriented jobs performed by administrative as well as clinical staff in this scenario that can impact the core functional job being studied.

There are also emotional jobs and spiritual jobs that need to be identified. These will emerge during the course of conversation, during interviews and/or panel discussions with individuals who are performing the job(s) being studied. The key challenge is to develop a consistent taxonomy for describing these types of jobs to ensure the data can be aligned with the other jobs and measurements in support of the downstream analytics.

Examples of emotional and spiritual jobs:

- Avoid feeling frustrated
- Be perceived as competent
- Feel valued in the eyes of G-d
- Being able to make a difference in the lives of others
- Feel that my life has made a difference

5.4.3. Preparation. Once the landscape for the study has been mapped out, the next step involves researching the core functional jobs that will be studied. This research begins by interviewing subject matter experts (SMEs) - ideally individuals who have performed or who currently performing the core functional job being studied. Additional materials for research may include existing standard operating procedures (SOPs), policy manuals, and guidelines. These materials provide contextual understanding for a given job and provide insights on the constraints that impact the job being studied. Observation is also helpful, especially if this is a job that is new to the research team since this provides a “walking in their shoes” perspective.

Interviews with SMEs are an important part of the preparation since these folks can provide initial guidance on mapping out the steps in a given Job. They can also provide input on the other functional jobs as well as the emotional and spiritual

jobs that are all part of building the complete reference model for a given job. These interviews also provide a practice run with the actual job executors and an initial set of desired outcomes can be captured as a baseline for the study.

Additional preparations involve scheduling resources for interviews, establishing a script and identifying the right metadata to capture for profiling the study participants. The metadata is a critical piece and one that requires careful planning to ensure it aligns to the metamodels previously defined in step 1. The metadata fields drive the dimensional analysis once the quantitative data has been captured. These data fields should be derived from a metadata repository to ensure the downstream analysis can be extended across job studies and across the domains.

5.4.4. Confirmation. This step is critical to ensure there is up-front alignment with the sponsors of the study as well as for managing expectations amongst the research team and study participants. Ideally, the confirmation should be conducted with a steering committee which includes sponsors, key decision makers, an individual who is responsible for the data architecture as well as appropriate data stewards and information product managers. This will help to ensure data integrity is in place up front and throughout the study.

Key artifacts that should be reviewed include (but are not limited to):

- The Domain Model
- Job Ontology Model
- Job Reference Model
- The Interview Schedule and list of participants

5.4.5. Execution. The execution of the study involves 2 phases – a Qualitative Study and a Quantitative Study. The Qualitative Study is used to capture the desired outcomes, which are then used to build the survey for capturing the value measurements as part of the Quantitative Study. Throughout each of these phases, there may be cycles of monitoring and modification to clean data, adjust findings and provide continual improvement to ensure an optimal set of results are obtained from the study.

5.4.5.1. Qualitative Study. This follows along the lines of traditional market research with customers, where interviews are setup with designated job executors as one-on-one meetings, panel discussions, and/or phone calls.

Notes that are captured during these meetings should be contextually relevant to the type of job (core functional job, non-core functional job, supporting job, emotional job, and/or spiritual job). Care should be taken to ensure the notes indicate direction for each desired outcome, that is whether the outcome metric is measurable (i.e. shows an increase or decrease) with respect to its current state. Direction is fairly straightforward to determine when dealing with quantifiable outcomes (such as time, effort, or cost), however, it becomes more challenging when dealing with qualified outcomes (such as risk, error, interruption, availability, fatigue, mobility or adverse reaction), which are not always quantifiable from the perspective of the person executing a given job.

As the notes are captured, it's important to transform them into desired outcomes. Along the way, new information may warrant updates to the Job Reference Model based on findings to ensure that outcomes are aligned with individual steps in the Job Reference Model. Additional data alignment includes identifying other Functional jobs that directly or indirectly impact the Core Functional Job being

studied. Consider upstream supply chain Jobs in a pharmacy that are being performed by one or more non-core functional job executors. If the purchasing department does not ensure that inventory is maintained properly, if the formulary manager fails to stock appropriate drugs, or the pharmaceutical company fails to meet quality and safety standards in manufacturing, a pharmacist may pull drugs off the shelf that are expired, or worse, a nurse may administer medication that may either not be effective or may even cause harm to the patient. Similarly, if medical record data quality is not properly controlled or maintained, the value of electronic medical records becomes diminished, or worse, a doctor may not make the right diagnosis or treatment recommendation. Of critical importance is to identify any Emotional or Spiritual Jobs and align them to individual steps and/or jobs in the Reference Model. These relationships will prove important during later analysis.

Once all the data has been collected and the alignment and relationships between them have been documented, the desired outcomes need to go through a rigorous process of cleansing to ensure the data:

- Conforms to an agreed upon taxonomy;
- Conforms to a consistent grammar (i.e. it contains action, direction, context, and is focused by relevant examples) ;
- Conforms to a proper statement syntax (i.e. ordering of the outcome statement components is important to ensure consistent interpretation) ;
- Ensures uniqueness amongst the desired outcomes for each step;
- Creates concise statements by consolidating desired outcomes which are “variations on theme” ;
- Eliminates bias in statements that discuss a particular solution, product, or judgment.

The last step involves verifying the cleansed listing of Desired Outcomes with the study participants to ensure the list accurately portrays the information provided during the interviews with the study participants. Once their feedback has been incorporated into the list of desired outcomes, the survey tool may be built.

5.4.5.2. Quantitative Study. The survey is prepared by developing profiling questions, providing a script for the survey executor to read and identifying a target audience for the survey. The same care with respect to metadata should be used when preparing the profiling questions as was used in preparing the Qualitative Study. The target audiences should be a statistically appropriate sample of the executors for a given job.

In order to limit survey error (due to subject fatigue) separate surveys should be prepared and presented to different subsets of the sample population by varying the scope and content of the questions in the instrument. The statements should be directed to the specific job executor, for example, if outcomes were captured from both pediatric nurses and adult care nurses, we would want to evaluate these outcomes individually and collectively to assess areas of convergence and divergence. The survey should measure two metrics for each job the executor is trying to do: first, the importance of the desired outcome; and second, the current level of satisfaction with the desired outcome.

The survey is best conducted by a third party to ensure anonymity. Profiling information (including but not limited to job title, number of years on the job,

degree, gender, ethnicity, location, etc.) should be captured from each respondent. Surveys can be conducted via phone, web and/or written response.

Once the survey data has been captured it requires appropriate lifecycle management according to best practices (Lee, Pipino, Funk, & Wang, 2006).

5.4.5.3. *Data Analysis*. The last task is to perform additional alignment with other data associated with the job study as well as prior studies. Formal pattern extraction is performed across multiple dimensions and assimilated into a consolidated abstraction of themes to guide opportunities for improving processes, products, and services. Below is a baseline set of 4 areas for guiding the data analysis for a given study:

- For the Core Functional Job;
- Across Jobs within a Job Study;
- Across other Jobs in a Domain;
- Across other Metrics.

5.4.6. *Conclude*. This is the last step in the VDM methodology used to extract themes and focus only on the patterns impacting those desired outcomes with highest opportunity (i.e. those where the Importance of a desired outcome is High and the Satisfaction is Low). In order to visualize and easily identify the highest opportunity candidates, each desired outcome is located on a scatter plot of Importance versus Satisfaction. Figure 7 illustrates this mapping for functional, emotional, and spiritual outcomes with respect to the assessment of Importance and Satisfaction from a patient perspective. Note that the greatest opportunities for improving patient outcomes are in the “under served” region in the lower right-hand corner of the chart. Those outcomes that fall within the “over served” area in the upper left-hand corner of the chart provide opportunities for improving efficiency as there is likely to be excess effort, resource utilization and cost associated with the overly high levels of satisfaction relative to the lower level of importance given by patients to these desired outcomes. The findings are then presented to the steering committee for review along with recommendations on how to align PDM and RDM metrics with the VDM findings.

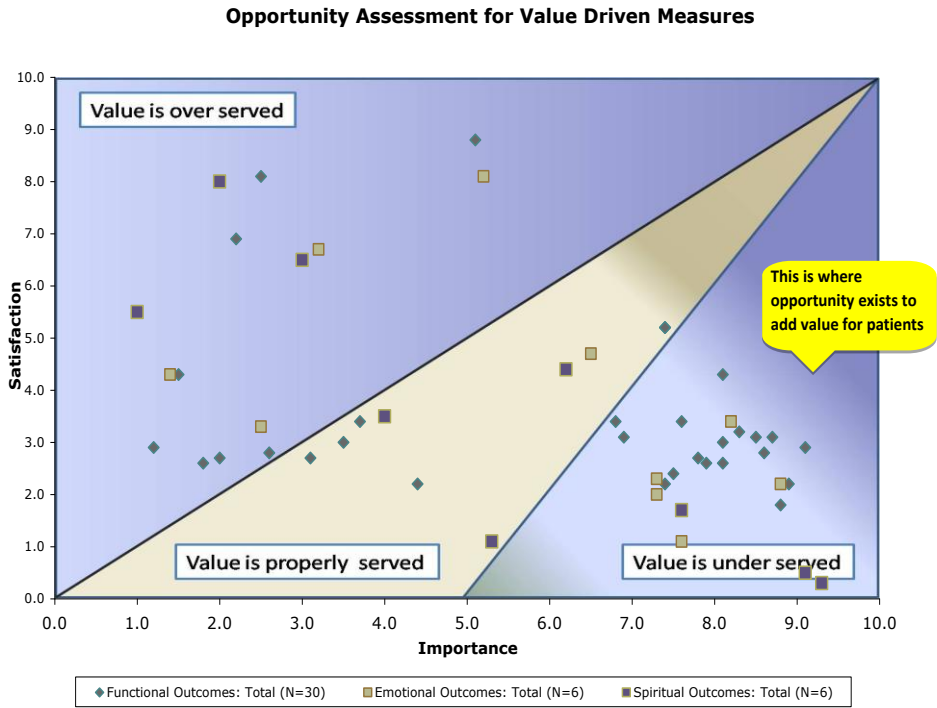


Fig. 7 Scatter Plot showing Value Opportunities (adapted from Ulwick, 2005)

6. CONCLUDING THOUGHTS

Revisiting Berwick’s automobile analogy for describing our healthcare system, we need to embark on designing a car that can adapt to the future needs of healthcare at an affordable cost that addresses all the measures of quality. As a complete ground-up, VDM-based redesign of the automobile is a Herculean task, we propose instead an initial redesign of the automobile dashboard so that it can be simplified (by aligning all the metrics) and we propose a new navigation system (using patient-driven outcomes) to provide us with a steady compass as we chart a course to guide the transformation of our healthcare system.

Our current system of healthcare is one of the most complex systems ever created from both business and care delivery viewpoints. As such, all of the Execution Methodologies (e.g. LEAN, TQM, EBM, CER, peer- and case-review, templates and protocols) discussed in Section 2.2 have key roles to play in assuring efficient and effective processes performed by process-workers and providers in order to deliver consistent, high-benefit outcomes to patients. The foundations for each of these Execution Methodologies are firmly rooted in the H-DA and therefore squarely focus on applying the scientific method to the evidence and observations at hand such that the best possible medical decisions can be made. So long as these Execution Methodologies are only applied to PDM and RDM, they will only be suited to address the functional jobs that

patients need to get done, which jobs represents only one part of the continuum-of-care. There are many other facets of care beyond the purely functional, and these emotional and spiritual facets can be as important to the patient as any functional outcome.

PDM leverages its roots in manufacturing to focus on measuring and implementing process change in “near real time”. PDM empowers individual process workers to identify waste and rapidly implement solutions to reduce that waste in a quantifiable manner. PDM has demonstrated the value that *fitness-for-purpose* can provide and that healthcare processes can be precisely managed. PDM has directly improved the efficiency and effectiveness of the healthcare system and indirectly thereby has improved the quality of outcomes for patients.

RDM shifts the focus from processes, waste in these processes and the role of the process worker in minimizing waste to consistent, high-benefit outcomes engineered by providers. But, in its focus on provider-chosen outcomes and process changes, RDM falls short of capturing more meaningful and strategic measures, namely those deemed valuable by the patients themselves. The successes of RDM have demonstrated that performance can be measured and improved from the provider viewpoint in a manner complementary to the improvements realized by PDM. With specifications set by the provider, *fitness-for-purpose* in RDM can readily be optimized.

Since RDM and PDM are both focused on *fitness-for-purpose*, they have fallen short in overcoming the challenges of *fitness-of-purpose* and systemic integration, especially where management and clinical performance overlap. Further, neither PDM nor RDM transitions smoothly across silos, whether organizational or functional. The stability of RDM and PDM measures is inconsistent since they have unequal emphasis on process and results, respectively. The introduction of patient-centric measurement (Bethell, 2000) is an implicit admission of the known limitations of RDM and PDM methods.

VDM is introduced to provide a model (see Figure 8) for establishing a stable foundation for measuring the gaps in integration across the continuum-of-care and the gaps in the horizontal flow of products and services throughout the healthcare value network in order to drive continuous improvement and innovation in healthcare without sacrificing the desired outcomes of patients. Since Value-Driven Measures are captured from patients using a disciplined methodology, the metrics can be mapped across the continuum-of-care, therefore building much needed alignment into PDM and RDM. In addition, VDM metrics are based on statistically valid measurements of outcomes based on a modeled view of activities (Jobs) that patients are trying to perform. This abstraction provides insulation from any given solution-oriented technology, and has a temporal stability that allows these measures to remain in use over relatively long periods of time.

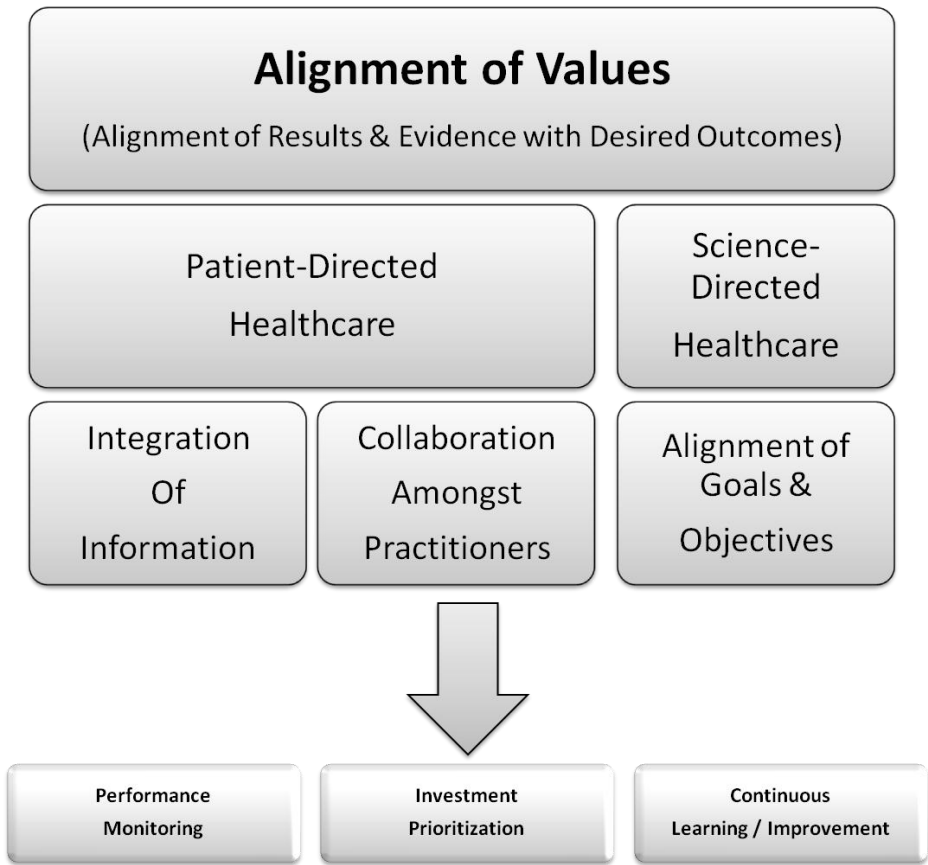


Fig. 8. Model for Driving Improvement in Healthcare via VDM

Within VDM, there is a need to consider the impact of non-core functional jobs which directly and indirectly impact the outcomes in the core functional job. To provide a more holistic understanding of the patient experience, we need to understand the relationships between the emotional and spiritual, as well as the functional, jobs.

We intend to demonstrate the unifying ability of VDM by capturing data from projects in hospitals, clinics, nursing homes and alternate care facilities (including the home). The lead author has participated in preliminary, proprietary VDM studies which have been carried out in hospital and clinic settings. The success of these preliminary studies prompted us to genericize, elaborate and extend the proprietary VDM methodology so as to spur research and interest in VDM as a complement to PDM and RDM. We envision that as more studies are performed across multiple stakeholder constituencies, the data collected along the continuum-of-care will evolve into a more complete understanding of the modifications necessary to design Berwick’s automobile for optimal performance today and in the future.

6.1. BEYOND PROCESS WORKERS, PROVIDERS AND PATIENTS

In order to create a truly holistic framework of measurement, we need to connect value as determined by the patient to all stakeholders who contribute to the creation of that value. We have mentioned the payer as one of these stakeholders; we have not mentioned the policy-maker as one. In this context we are focusing on the regulatory bodies that are focused on compliance-driven measures. A full exploration of the impact of the policy-maker and compliance on value as determined by the patient deserves its own paper. Measuring compliance is not straightforward due to the way that regulations are authored and implemented as guidelines subject to interpretation. This ambiguity is intentional as the practical realities of creating a completely unambiguous specification are beyond the ability of regulatory agencies. Transposing regulatory guidelines into rules and metadata is challenging, however, once done, it is possible to align policy-makers, providers, payers and process workers with patients. Doing so is critical to ensuring that VDM is fully adopted and that the gaps in value delivery are properly acted upon to improve the products and services delivered to patients.

6.2. CONNECTING THE DOTS...

Porter's message (Porter, 2009) drives to the heart of the issue with healthcare today. As Porter states, the central focus needs to be delivering value to the patient. The desired outcomes that patients want to achieve will provide a consistent baseline for measuring the value that healthcare needs to deliver. These metrics will provide the foundation to unify the fragmented healthcare system. Once healthcare practitioners have the right metrics, they can drive value all the way back through the value delivery network to ensure that each of the participants in this network are achieving their desired outcomes. This approach will allow us to build a fully meshed network of customers and metrics, which will provide a measureable baseline for designing a value-driven model for healthcare. Once we have the right metrics, we can align the products and services across providers and payers to ensure that each is compliant and efficient in the delivery of their value contributions. VDM will help guide the growing movement of patient-directed healthcare so that it can productively co-exist and evolve with science-directed healthcare (see Figure 8), while only causing minimal disruption to the desired outcomes for patients. The VDM methodology can be extended to capture desired outcomes for providers and payers across the value networks within healthcare. Extending the analysis to include all stakeholders will provide us with a progressive model to improve healthcare in a systematic manner by first aligning the data used to measure value, results, and evidence to guide integration of information, collaboration amongst providers, and drive improvements in the products, processes, and services delivered by the providers throughout an increasingly distributed healthcare continuum.

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